

[PLACEHOLDER] Developement of a Segmented Communication Protocol for Sensor Data Transmission Using LoRa[®] Technology

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Abstract—The abstract goes here...

Index Terms—

I. INTRODUCTION

LOW-POWER, low-cost and long-range wireless communication technologies are becoming increasingly important in the context of the Internet of Things (IoT). Applications such as environmental monitoring, precision agriculture or Unmanned Aerial Vehicles (UAVs), require reliable data transmission over extended distances with strict power and hardware constraints.

A range of different communication standards, including ZigBee, Sigfox and Long Range (LoRa[®]) have been adopted to address these needs, and LoRa[®], in particular, has emerged as the leading choice for low-power and long-range deployments thanks to its robust modulation technique and open network architecture. However, conventional solutions based on LoRa[®] (particularly those following the Long Range Wide Area Network (LoRaWAN[®]) specifications) are optimized for small, non-frequent payloads.

This paper presents the design and implementation of a custom communication protocol for LoRa[®], engineered to enable the transmission of segmented large data payloads over long distances. The proposed system has been deployed on a sounding rocket built by the Aurora Rocketry team in the University of Bologna, demonstrating its applicability to highly dynamic systems, that require continuous transmission of large amount of telemetry data with high reliability and low power consumption in a point-to-point configuration. This approach is generalizable to a wide range of IoT applications requiring extended data capacity over extended distances and high power efficiency, and possibly to multi-agent systems that require the transmission of structured data between the agents. [1].

II. STATE OF THE ART

III. PROPOSED SYSTEM

The proposed system uses a point-to-point, unidirectional architecture, with a transmitter mounted on the rocket's flight computer and a receiver mounted inside the ground station.

The LoRa modules used in this implementation had a packet size limit of 200 Bytes.

The base LoRa[®] packet generated by the adopted modules is structured as follows:

- **Physical Header (3 Bytes):**
 - High Address (1 Byte)
 - Low Address (1 Byte)
 - Communication Channel ID (1 Byte)
- **Payload (up to 194 Bytes)**
- **Footer (2–3 Bytes):**
 - Cyclic Redundancy Check (CRC) (2 Bytes)
 - Received Signal Strength Indicator (RSSI) (1 Byte, optional, automatically added by the LoRa[®] module and enabled in this implementation for link quality monitoring)

In addition to the 3-byte physical header defined by the LoRa module, a custom 11-byte logical header is prepended to each payload to manage segmentation. The extension enables transmission of larger data structures by segmenting them into smaller chunks. The custom header is structured as follows:

- **Packet Number (2 Bytes):** Unique identifier of the segmented payload sequence.
- **Total Chunks (1 Byte):** Total number of chunks the payload has been divided into.
- **Chunk Index (1 Byte):** Progressive index identifying the current chunk within the sequence.
- **Chunk Size (2 Bytes):** Effective size (Bytes) of the current chunk.
- **Payload Size (2 Bytes):** Size (Bytes) of the original payload before segmentation.
- **Timestamp (4 Bytes):** Time reference associated with the packet, useful for performance analysis and synchronization.
- **Protocol Version (1 Byte):** Reserved field for future protocol revisions, ensuring backward compatibility.

The final produced packet is represented in Fig. 1.

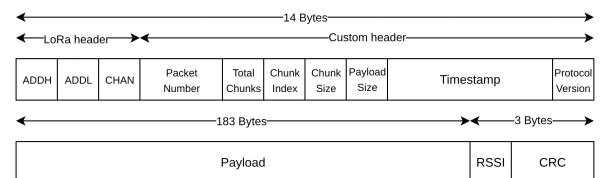


Fig. 1. Transmitted packet schematic representation.

TABLE I
CUSTOM PROTOCOL PACKET STRUCTURE

Field	Size (Bytes)	Description
Physical Header		
ADDL	1	Low address (LoRa module physical address)
ADDH	1	High address (LoRa module physical address)
CHAN	1	Channel ID selected on the LoRa module
Custom Logical Header		
Packet Number	2	Unique identifier of the segmented payload sequence
Total Chunks	1	Number of chunks in the segmented payload
Chunk Index	1	Index of the current chunk in the sequence
Chunk Size	2	Actual size (Bytes) of the transmitted chunk
Payload Size	2	Size (Bytes) of the original payload before segmentation
Timestamp	4	Time reference (ms since transmission start)
Protocol Version	1	Reserved byte for protocol retro-compatibility
Payload		
Payload	183	Telemetry data content
Footer		
RSSI (optional)	1	Received Signal Strength Indicator (added by the LoRa module)
CRC	2	Cyclic Redundancy Check for error detection

This protocol enables robust packet segmentation and allows a flexible implementation to send any kind of data structure through LoRa[®]. The packet sequence reconstruction is also customizable.

IV. IMPLEMENTATION

V. TESTING

VI. CONCLUSION

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APPENDIX A

TITLE OF APPENDIX ONE

Appendix one text goes here.

APPENDIX B

Appendix two text goes here.

ACKNOWLEDGMENT

The authors would like to thank...

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- [1] M. Radeta, J. Pestana, P. Abreu, R. Freitas, F. Silva, D. Vieira, R. Prieto, M. Fernandez, F. Alves, T. Dellinger, S. Neves, and E. Delory, "Triton—open telemetry and location estimation for marine monitoring based on iot and lora," *IEEE Journal of Oceanic Engineering*, vol. 50, no. 2, pp. 1244–1258, 2025.